

Stochastic Calculus Notes

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Contents

1	Random Walks	2
2	Stock Price Process	4
3	Ito Calculus	6

1 Random Walks

1. Let ω_i denote the outcome of the i th coin toss. We define,

$$\mathbb{X}_i(\omega_i) = \begin{cases} +1, & \omega_i = H \\ -1, & \omega_i = T \end{cases}$$

2. Also define,

$$\mathbb{M}_n = \sum_{i=1}^n \mathbb{X}_i$$

3. Note that, $-n \leq \mathbb{M}_n \leq n$. Also, $\mathbb{M}_n$ and n must have the same parity.
4. Probability that the walker is at $\mathbb{M}_n = k$ after n steps is

$$\mathbb{P}\{\mathbb{M}_n = k\} = p_{n,k} = \frac{1}{2^n} \binom{n}{\frac{n+k}{2}}$$

5. Probability of return to the original position, $k = 0$ after $n = 2\nu$ steps is

$$\mathbb{P}\{\mathbb{M}_{2\nu} = 0\} = \mu_{2\nu} = \binom{2\nu}{\nu} \frac{1}{2^{2\nu}}$$

6. Number of paths from $(0, 0)$ to (n, x) are,

$$N_{n,x} = \binom{n}{\frac{n+x}{2}}$$

7. **Reflection Principle:** The number of paths from (a, α) to (b, β) with $\alpha, \beta > 0$ which touch or cross the t-axis at least once is equal to the total number of paths from $(a, -\alpha)$ to (b, β) .
8. Number of paths from $(0, 0)$ to (n, x) that stay strictly above or below the t-axis can be calculated by taking the first step to $(1, 1)$ and then using the Reflection Principle. In our case, the paths come out to be

$$N_{n-1,x-1} - N_{n-1,x+1} = \frac{x}{n} N_{n,x}$$

9. Unrelatedly, but obviously, and clearly, $\sum_{k=-n}^n N_{n,k} = 2^n$, which is the total possible paths in n time steps.
10. Probability of walker returning to $k = 0$ for the first time in $n = 2\nu$ steps, $f_{2\nu}$ can be calculated from the following relation.

$$\mu_{2\nu} = f_{2\nu}\mu_0 + f_{2\nu-2}\mu_2 + \dots + f_2\mu_{2\nu-2}$$

11. We can iteratively solve this but let's go for a much more elegant method. For the sake of precision,

$$f_{2\nu} = \mathbb{P}\{\mathbb{M}_1 \neq 0, \mathbb{M}_2 \neq 0, \dots, \mathbb{M}_{2\nu-1} \neq 0, \mathbb{M}_{2\nu} = 0\}$$

12. We now prove the following result, which would be instrumental in getting a closed form solution for $f_{2\nu}$. Consider the result:

$$\mathbb{P}\{\mathbb{M}_1 \neq 0, \mathbb{M}_2 \neq 0, \dots, \mathbb{M}_{2\nu} \neq 0\} = \mathbb{P}\{\mathbb{M}_{2\nu} = 0\} = \mu_{2\nu}$$

Proof. Total number of paths that stay above the t-axis but can be anywhere (> 0) at $n = 2\nu$ is

$$\sum_{l=1}^{\nu} (N_{2\nu-1,2l-1} - N_{2\nu-1,2l+1}) = N_{2\nu-1,1}$$

Also, the probability of getting such paths is

$$p_{2\nu-1,1} = (N_{2\nu-1,1}) \frac{1}{2^{2\nu}} = \binom{2\nu-1}{\nu} \frac{1}{2^{2\nu}}$$

To get the expression for our result, we must double the above quantity to account for the paths that stay strictly below the t-axis. Thus,

$$2p_{2\nu-1,1} = \frac{(2\nu)!}{\nu!\nu!} \frac{1}{2^\nu} = \mu_{2\nu}$$

□

13. Again consider, $f_{2\nu} = \mathbb{P}\{\mathbb{M}_1 \neq 0, \mathbb{M}_2 \neq 0, \dots, \mathbb{M}_{2\nu-1} \neq 0, \mathbb{M}_{2\nu} = 0\}$. All length 2ν paths which satisfy $\{\mathbb{M}_1 \neq 0, \mathbb{M}_2 \neq 0, \dots, \mathbb{M}_{2\nu-2} \neq 0\}$ are $4(2^{2\nu-2}\mu_{2\nu-2}) = 2^{2\nu}\mu_{2\nu-2}$, but this is equal to sum of paths that return to $k = 0$ at 2ν and those that do not. Thus,

$$2^{2\nu}\mu_{2\nu-2} = 2^{2\nu}f_{2\nu} + 2^{2\nu}\mu_{2\nu}$$

14. From the above, we get,

$$f_{2\nu} = \mu_{2\nu-2} - \mu_{2\nu}$$

Also, trivially,

$$f_{2\nu} = \frac{1}{2\nu-1}\mu_{2\nu}$$

15. Probability of the walker returning back to 0 for the last time at $t = 2l$ in an $n = 2\nu$ step walk is,

$$\alpha_{2l, 2\nu} = \mu_{2l} \cdot \mu_{2\nu-2l}$$

16. If ω denotes the sequence of outcomes of a coin toss experiment, and $n_H(\omega)$ and $n_T(\omega)$ the number of Heads and Tails in that sequence, then

$$\mathbb{P}(\omega) = p^{n_H(\omega)} q^{n_T(\omega)}$$

- 17.

$$\mathbb{E}\{\mathbb{M}_n\} = n(p - q)$$

18. **Markov Property:** The value of a random variable *only* depends on the previous step. $\mathbb{M}_n$ is a Markov Process.

Proof.

$$\mathbb{P}\{\mathbb{M}_n = k \mid \mathbb{M}_{n-1} = a_{n-1}, \dots, \mathbb{M}_1 = a_1\} = \frac{\mathbb{P}\{\mathbb{M}_n = k, \mathbb{M}_{n-1} = a_{n-1}, \dots, \mathbb{M}_1 = a_1\}}{\mathbb{P}\{\mathbb{M}_{n-1} = a_{n-1}, \dots, \mathbb{M}_1 = a_1\}}$$

This evaluates to $\frac{\frac{1}{2^n}}{\frac{1}{2^{n-1}}} = \frac{1}{2}$, but the following quantity also evaluates to the same. Thus,

$$\mathbb{P}\{\mathbb{M}_n = k \mid \mathbb{M}_{n-1} = a_{n-1}, \dots, \mathbb{M}_1 = a_1\} = \mathbb{P}\{\mathbb{M}_n = k \mid \mathbb{M}_{n-1} = a_{n-1}\}$$

□

19. **Conditional Expectation**

$$\mathbb{E}[\mathbb{X}](\omega_1, \omega_2, \dots, \omega_n) = \sum_{\substack{\omega \in \Omega_N \\ \text{s.t. } \omega_1, \omega_2, \dots, \omega_n \text{ fixed} \\ \text{sum over } \omega_{n+1}, \omega_{n+2}, \dots, \omega_N}} \mathbb{X}(\omega) \mathbb{P}\{\omega \mid \omega_1, \omega_2, \dots, \omega_n\}$$

20. **Martingale Property:**

$$\mathbb{E}[\mathbb{M}_n \mid \mathcal{F}_m] = \mathbb{M}_m, \quad m \leq n$$

21. **Sub-Martingale:** $\mathbb{E}[\mathbb{M}_n \mid \mathcal{F}_m] \geq \mathbb{M}_m$

22. **Super-Martingale:** $\mathbb{E}[\mathbb{M}_n \mid \mathcal{F}_m] \leq \mathbb{M}_m$

23. $\mathbb{M}_n$ is a martingale. As,

$$\mathbb{E}[\mathbb{M}_n](\omega_1, \omega_2, \dots, \omega_m) = \mathbb{M}_m(\omega_1, \omega_2, \dots, \omega_m), \quad \text{for } m < n$$

2 Stock Price Process

1. **Binomial Tree Model:** An N -step model where the stock price can either increase or decrease by a constant up or down factor:

$$\mathbb{S}_n = u^{n_H(\omega_1\omega_2\ldots\omega_n)} d^{n_T(\omega_1\omega_2\ldots\omega_n)} \mathbb{S}_0.$$

For a symmetric model, we have $u = 1/d$, and thus

$$\mathbb{S}_n = u^{M_n(\omega_1\omega_2\ldots\omega_n)} \mathbb{S}_0.$$

2. Properties of this Stock Price Process

- (a) $\mathbb{S}_n$ is an adapted stochastic process.
- (b) $\mathbb{S}_n$ is a Markov process.
- (c) $\mathbb{S}_n$ is a martingale under specific probabilities called the *risk-neutral probabilities*.
- (d) The conditions $\tilde{p}u + \tilde{q}d = 1 + r$ and $\tilde{p} + \tilde{q} = 1$ yield

$$\tilde{p} = \frac{1 + r - d}{u - d}, \quad \tilde{q} = \frac{u - (1 + r)}{u - d}.$$

- (e) The discounted stock price process, defined below, is a martingale under the risk-neutral measure:

$$\mathbb{Y}_n(\omega_1\omega_2\ldots\omega_n) = \frac{\mathbb{S}_n(\omega_1\omega_2\ldots\omega_n)}{(1 + r)^n}.$$

3. **Hedging Portfolio:** Suppose a seller of a European call option hedges their risk by purchasing the underlying stock and investing the remainder in the money market. To be fully hedged, the value of the portfolio must equal the payoff of the option in all possible states; otherwise, there is a possibility of arbitrage.
4. The payoff of the call option is $\mathbb{V}_N = \max(\mathbb{S}_N - K, 0)$, and the portfolio evolves as

$$\mathbb{X}_n(\omega_1\ldots\omega_n) = \Delta_{n-1}(\omega_1\ldots\omega_{n-1})\mathbb{S}_n(\omega_1\ldots\omega_n) + (1 + r)(\mathbb{X}_{n-1}(\omega_1\ldots\omega_{n-1}) - \Delta_{n-1}(\omega_1\ldots\omega_{n-1})\mathbb{S}_{n-1}(\omega_1\ldots\omega_{n-1})).$$

5. **For $N = 1$:** $\mathbb{X}_1(\omega_1) = (\mathbb{S}_1(\omega_1) - K)^+$, and

$$\Delta_0 = \frac{\mathbb{X}_1(H) - \mathbb{X}_1(T)}{\mathbb{S}_1(H) - \mathbb{S}_1(T)},$$

$$\mathbb{X}_0 = \frac{1}{1 + r} [\tilde{p}\mathbb{X}_1(H) + \tilde{q}\mathbb{X}_1(T)] = \tilde{\mathbb{E}}\left(\frac{\mathbb{X}_1}{1 + r}\right).$$

6. In general,

$$\Delta_n(\omega_1\ldots\omega_n) = \frac{\mathbb{X}_{n+1}(\omega_1\ldots\omega_n H) - \mathbb{X}_{n+1}(\omega_1\ldots\omega_n T)}{\mathbb{S}_{n+1}(\omega_1\ldots\omega_n H) - \mathbb{S}_{n+1}(\omega_1\ldots\omega_n T)},$$

$$\mathbb{X}_n(\omega_1\ldots\omega_n) = \frac{1}{1 + r} [\tilde{p}\mathbb{X}_{n+1}(\omega_1\ldots\omega_n H) + \tilde{q}\mathbb{X}_{n+1}(\omega_1\ldots\omega_n T)].$$

7. Wiener Process

- (a) It is easy to verify that $\mathbb{E}[\mathbb{M}_n] = 0$ and $\mathbb{E}[\mathbb{M}_n^2] = n$.
- (b) **Scaled Random Walk:** We define

$$\mathbb{W}^{(p)}(t) = \frac{1}{\sqrt{p}} \mathbb{M}_{pt},$$

where

$$\mathbb{M}_{pt} = \begin{cases} \mathbb{M}_{pt} & \text{if } pt \text{ is an integer,} \\ \text{linear interpolation} & \text{otherwise.} \end{cases}$$

- (c) The increment of the random walk is $\mathbb{M}_n - \mathbb{M}_m = \sum_{i=m+1}^n \mathbb{X}_i$. Clearly, $\mathbb{E}[\mathbb{M}_n - \mathbb{M}_m] = 0$ and $\mathbb{E}[(\mathbb{M}_n - \mathbb{M}_m)^2] = n - m$.
- (d) The Wiener process is defined as the limit

$$\mathbb{W}(t) = \lim_{p \rightarrow \infty} \mathbb{W}^{(p)}(t) = \lim_{p \rightarrow \infty} \frac{1}{\sqrt{p}} \mathbb{M}_{pt}.$$

- (e) The process exhibits fractal (self-similar) behaviour and Hölder continuity. The scaling factor $\sqrt{p}$ is chosen so that the quadratic variation of the Wiener process equals t .
- (f) **Properties of the Wiener Process**
- i. $\mathbb{W}(t) \sim N(0, t)$.
 - ii. For $0 < t_1 < t_2 < \dots < t_n$, the increments $\mathbb{W}(t_i) - \mathbb{W}(t_{i-1})$ are independent.
 - iii. $\mathbb{W}(t_i) - \mathbb{W}(t_{i-1}) \sim N(0, t_i - t_{i-1})$.
- (g) Since $\mathbb{W}(t)$ is normally distributed, we have

$$\mathbb{P}\{\mathbb{W}(t) \leq \omega\} = \frac{1}{\sqrt{2\pi t}} \int_{-\infty}^{\omega} e^{-x^2/(2t)} dx.$$

8. Continuous Limit of the Binomial Asset Pricing Model

The scaled stock price is given by

$$\mathbb{S}_{pt} = u^{n_H} d^{n_T} \mathbb{S}_0 = u^{(pt + \mathbb{M}_{pt})/2} d^{(pt - \mathbb{M}_{pt})/2} \mathbb{S}_0.$$

Choosing u and d such that $ud = 1$ and, in the limit as $p \rightarrow \infty$,

$$u = 1 + \frac{\sigma}{\sqrt{p}}, \quad d = 1 - \frac{\sigma}{\sqrt{p}},$$

we obtain

$$\mathbb{S}_{pt} = \left(1 + \frac{\sigma}{\sqrt{p}}\right)^{(pt + \mathbb{M}_{pt})/2} \left(1 - \frac{\sigma}{\sqrt{p}}\right)^{(pt - \mathbb{M}_{pt})/2} \mathbb{S}_0.$$

Taking $p \rightarrow \infty$ yields

$$\lim_{p \rightarrow \infty} \mathbb{S}_{pt} = \mathbb{S}(t) = \mathbb{S}_0 \exp\left(\sigma \mathbb{W}(t) - \frac{1}{2} \sigma^2 t\right),$$

which is the *geometric Brownian motion*.

9. Recall that $\mathbb{E}[(\mathbb{W}(t_i) - \mathbb{W}(t_{i-1}))^2] = t_i - t_{i-1}$, which informally gives $\mathbb{E}[(\Delta \mathbb{W})^2] = \Delta t$, and hence $d\mathbb{W}^2 = dt$. Consequently, $d\mathbb{W}/dt \sim 1/\sqrt{dt}$, which diverges as $dt \rightarrow 0$; the Wiener process is therefore nowhere differentiable in the classical sense. To carry out calculus with $d\mathbb{W}$, we must develop a new framework known as *Itô calculus*.

3 Ito Calculus